

Section 5.5 Similarity and Diagonalization

Exercise 5.5.1

By computing the trace, determinant, and rank, show that A and B are not similar in each case.

d. $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} \quad B = \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$

$\text{Trace}(A; 5) \quad \text{Trace}(B; 4)$
 $\therefore \text{Trace}(A) \neq \text{Trace}(B)$

$\text{Det}(A) = 6 + 1 = 7 \quad \therefore \text{Det}(A) = \text{Det}(B)$

$\text{Det}(B) = 4 + 9 = 7$

$\text{Rank}(A) = \text{Rank}(B) = 2$

$\therefore \text{Rank, Det are equal}$
 Trace not equal

e. $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ -3 & 6 & -3 \end{bmatrix}$

$\text{Trace}(A = 2) \quad \text{Trace}(B = 2)$
 $\therefore \text{Trace equal}$

$\text{Rank}; A = 2 \quad B = 1$
 $A \neq B \text{ for RANK}$

$\text{Det}(A) = -2(1) + 0 \neq \frac{1}{-2+2+1} = 1$
 $-1(1) \Rightarrow \text{Det}(A) = 0$
 $1(1)$

$\text{Det}(B) = 1(-12+12) = 0 \quad \therefore \text{Det}(A) = \text{Det}(B) = 0$
 $-(2(6-6))$
 $+ (1)(-12+12)$

Ted:...

Exercise 5.5.4

In each case, decide whether the matrix A is diagonalizable. If so, find P such that $P^{-1}AP$ is diagonal.

c. $A = \begin{bmatrix} 3 & 1 & 6 \\ 2 & 1 & 0 \\ -1 & 0 & -3 \end{bmatrix}$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

1) Solve for $(A - \lambda I) = 0$

$$\Rightarrow \det \begin{bmatrix} 3-\lambda & 1 & 6 \\ 2 & 1-\lambda & 0 \\ -1 & 0 & -3-\lambda \end{bmatrix} = 0 \Rightarrow \left. \begin{aligned} & (3-\lambda) \left[(1-\lambda)(-3-\lambda) - 0 \right] \\ & - 1(2(-3-\lambda) - 0) \\ & + 6(0 - (-1)(1-\lambda)) \end{aligned} \right\} \Rightarrow -\lambda^3 + \lambda^2 + 5\lambda + 3 = 0$$

Characteristic polynomial

Factor it out;

$$(\lambda - 3)(\lambda + 1)^2 = 0$$

$\therefore \lambda = \{3, -1\}$

$\therefore$ can be diagonalized

used online calculator
as director allow...

Finding eigen vectors; plug λ back in

^ put into R-ref

$$\Rightarrow \text{for } \lambda = -1 \Rightarrow \begin{bmatrix} 4 & 1 & 6 \\ 2 & 2 & 0 \\ -1 & 0 & -2 \end{bmatrix} \Rightarrow \begin{bmatrix} 4 & 1 & 6 \\ 0 & -3 & 6 \\ 0 & 1 & -2 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 & 0 & -2 \\ 2 & -2 & 0 \\ 4 & 1 & 6 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 & 0 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

$R_1 = R_1 - 2R_2$
 $R_2 = R_2 + 4R_1$

SWAP R_1, R_2 $-R_1 = R_1$

$R_2 = 2R_1 - R_2$
 $R_3 = 4R_1 - R_3$

$\Rightarrow -x_1 - 2x_3 = 0$
 $x_1 = -2x_3 = 0$

Basically see how only one free variable, dimension of eigenspace = 1, algebraic $\neq$ 2

$\therefore$ matrix can not be diagonalized.

Exercise 5.5.7

Let λ be an eigenvalue of A with corresponding eigenvector x . If $B = P^{-1}AP$ is similar to A , show that $P^{-1}x$ is an eigenvector of B corresponding to λ .

Given

$$\textcircled{1} B = P^{-1}AP$$

$\textcircled{2}$ Show $P^{-1}x$ eigenvector of B corresponding to λ

$\textcircled{3}$ means $Ax = \lambda x$ and $\therefore x \neq 0$

With $B(P^{-1}x)$; sub-def to get

$$= P^{-1}AP(P^{-1}x) = P^{-1}A(P\bar{P}^{-1})x$$

$$= P^{-1}(Ax); \text{ use } \textcircled{3} \text{ to simplify to}$$

$$\bar{P}^{-1}(\lambda x) = \lambda(\bar{P}^{-1}x).$$

λ
constant!

B/c $x \neq 0$, $\bar{P}^{-1}$ is invertible matrix; product cannot be 0 vector!

Exercise 5.5.10

Let A be a diagonalizable $n \times n$ matrix with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ (including multiplicities). Show that:

a. $\det A = \lambda_1 \lambda_2 \dots \lambda_n$

Know given 'definite' of ~~the~~ diagonalizable matrix

then must be some $A = PDP^{-1}$

Where 'D' contains eigenvalues of 'A' along its main diagonal such that $\det(A) = \det(PDP^{-1})$

$$P(\lambda) = \det(A - \lambda I) = (\lambda_1 - \lambda)(\lambda_2 - \lambda) \dots (\lambda_n - \lambda)$$

B/c λ are roots of polynomial we can do λ

① for $\lambda = 0$

②

$$P(0) = \det(A)$$

for the factored ver we have

$$P(0) = \lambda_1 \lambda_2 \lambda_3 \dots \lambda_n$$

B/c Both are $P(0)$ we can ③

$$\text{say } \det(A) = \lambda_1 \lambda_2 \dots \lambda_n = P(0)$$

b. $\text{tr} A = \lambda_1 + \lambda_2 + \dots + \lambda_n$

By def; $P(\lambda) = \det(A - \lambda I)$

The expanded definition:

$$\textcircled{B} P(\lambda) = (-1)^n \lambda^n + (-1)^{n-1} \text{tr}(A) \lambda^{n-1} + \dots + \det(A)$$

det calculated algebraically

def characteristic polynomial as;

$$\textcircled{A} P(\lambda) = (\lambda_1 - \lambda) \dots (\lambda_n - \lambda) = (-1)^n \lambda^n + (-1)^{n-1} \text{tr}(A) \lambda^{n-1} + \dots + \det(A)$$

Then for Both $\textcircled{A}, \textcircled{B}$, divide by $(-1)^{n-1}$ we can prove

$$\text{tr}(A) = \lambda_1 + \lambda_2 + \dots + \lambda_n$$

Section 6.1 Examples and Basic Properties

Exercise 6.1.2

Are the following sets vector spaces with the indicated operations? If not, why not?

c. The set of all polynomials of degree ≤ 3 ; operations of P .

Yes; Recall closure properties;

- under $+$
- under $\cdot$
- under '0-vector'

contains 0-vector ✓
all previous polynomials
inherit the 3 basic
properties.

However degree ≤ 3 does NOT work as it
doesn't include 0 vector!

d. The set $\{1, x, x^2, \dots\}$; operations of P .

NO

1) Doesn't contain 0 vector! if it had 0 would
work but only has $x^0 = 1$ $x^1 = 1 \dots$

i. Because it fails one property it fails all of
them!

h. The set V of real numbers; usual operations.

True; for $V = \mathbb{R}$, we can obtain 0-vector with $0+0, -1+1, \dots$ etc,
adding 2 real numbers ($\mathbb{R}$) results in $\mathbb{R}$

any scalar multiple results in another of itself;

$$K \times C = \mathbb{R}$$

constants $\in \mathbb{R}$

There is no actual division
(Only way we can get imaginary numbers)
as square roots aren't part of vector space operation

i. Holds True

under all conditions

1) closure

2) additive (+/-)

3) associative / commutativity

4) Distributive

Exercise 6.1.6

In each case show that the condition $au + bv + cw = 0$ in V implies that $a = b = c = 0$.

b. $V = M_{22}$; $u = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $v = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $w = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ AKA prove linear ind.

1) Combine into 1 matrix;

$$a \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + c \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a+c & b+c \\ b+c & a-c \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

2) make SoE

$$\left| \begin{array}{l} a+c=0 \\ b+c=0 \\ a-c=0 \end{array} \right| \begin{array}{l} -2c=0, \\ c=0 \\ a=0 \\ b=0 \end{array}$$

$\therefore b/c$ all trivial solutions hold,

They are linear ind.]

c. $V = P$; $u = x^3 + x$, $v = x^2 + 1$, $w = x^3 - x^2 + x + 1$

Same thing here; combine all $\Rightarrow$

$$a(x^3 + x) + b(x^2 + 1) + c(x^3 - x^2 + x + 1)$$

2) Group by x^n .

$$\Rightarrow a+c(x^3) + (b-c)x^2 + (a+c)x + (b+c) = 0$$

3) make SoE $\Rightarrow$ notice all $= 0$ for arb x , same as prev. problem!

$\Rightarrow \therefore$ linear ind.]

Exercise 6.1.9

Show that $-0 = 0$ in any vector space. Cite all axioms used.

$$\begin{aligned} -0 &= -0 + 0 \quad (0 \text{ additive identity}) \\ &= 0 \quad (\text{property of } \mathbb{A}) \end{aligned}$$

Combining Both we get axiom on inverse addition and identity of the element addition.

$$\therefore -0 = 0$$

Exercise 6.1.10

Show that the zero vector 0 is uniquely determined by the property in axiom A4.

like lost problem;

0 vector defined by

$$\underline{\vec{0} = -\vec{0} + \vec{0}}$$

Because of this; even if we add another $\vec{0}$ to the

existing one; the definition still holds;

1) commutative holds

2) closed

shows $\vec{0} = -\vec{0}$ (as per lost problem)

Exercise 6.1.11

Given a vector v , show that its negative $-v$ is uniquely determined by the property in axiom A5.

$$\left. \begin{array}{l} \text{2 vectors} \\ v + w_1 = 0 \\ v + w_2 = 0 \end{array} \right\} \Rightarrow \text{show } w_1 = w_2$$

By additive property know $w_1 = w_1 + 0$;
Sub $\nearrow$ into given;
 $w_1 = w_1 + (v + w_2)$

with associative property we get;
 $w_1 = (w_1 + v) + w_2$

$\Downarrow$
commutative property set

$$w_1 = (v + w_1) + w_2$$

Simplify Simplify

$$\Rightarrow w_1 = 0 + w_2$$

$$\text{or } w_1 = w_2 //$$